

Jinning LIANG

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RESEARCH INTERESTS

Keywords: galaxy formation theory, galaxy-halo connection, AGN feedback, BH subgrid physics, galaxy morphology, galactic dynamics, galactic chemical evolution, cosmology, numerical simulation, semi-analytical model

EDUCATION

The Kavli Institute for Astronomy and Astrophysics, Peking University <i>Ph.D. student in Astrophysics, Advisor: Prof. Fangzhou Jiang</i>	Sep 2025 - Sep 2029
The Institute for Computational Cosmology, Durham University <i>STFC-funded Ph.D. student in Physics, Advisor: Prof. Cedric Lacey</i>	Oct 2023 - Mar 2025 Oct 2023 - Sep 2024 Mar 2025
<i>M.Sc. in Physics received</i>	
School of Physics and Technology, Wuhan University <i>Bachelor of Science in Physics</i>	Sep 2019 - Jun 2023
➤ Astronomy Class (Selected from pool of 280 students due to outstanding performance and enthusiasm for astronomy)	Sep 2020 - Jun 2023
➤ GPA: 3.80/4.00; 90.15/100; Ranking: 1/196	

PUBLICATIONS (* Denotes co-first author, † Denotes corresponding author, Total Citation: 57, h-index: 6)

- [1] **Jinning Liang**[†], F. Jiang[†] et al., *AGN and IMBH in COLIBRE dwarf galaxies at $z=0$* , In prep.
- [2] **Jinning Liang**[†], F. Jiang[†] et al., *Galaxy size evolution in observations, simulations, and models*, To be submitted.
- [3] B. S. Kalita[†] et al. (including **Jinning Liang**), *SCALES. II. The Internal Composition of Giant Star-Forming Clumps at $z = 1 - 4$* , Submitted to ApJ, 2026 [[2610.03867](#)]
- [4] B. S. Kalita[†] et al. (including **Jinning Liang**), *SCALES. I. Bridging lensed Clump – Giant Clump Scales in Lensed Galaxies at $z = 1 - 4$* , Submitted to ApJ, 2026 [[2610.03866](#)]
- [5] J. Luo, F. Jiang[†], **Jinning Liang**, et al., *Ubiquitous nuclear disk and bars embedded in massive galaxies in the cosmic morning*, Submitted to ApJL, 2026 [[2609.27460](#)]
- [6] **Jinning Liang**, C. G. Lacey[†] et al., *Simulating AGN wind feedback with variable feedback efficiencies in idealised disc galaxies*, Submitted to MNRAS, 2026 [[2603.12322](#)]
- [7] **Jinning Liang**, F. Jiang[†], H.J. Mo, et al., *Connection between galaxy morphology and dark-matter halo structure II: predicting disk structure from dark-matter halo properties*, ApJ **1003** (2026) 12, 2026 [[2512.13822](#)]
- [8] F. Jiang^{†,*}, **Jinning Liang**^{*} et al., *Formation and Environmental Context of Giant Bulgeless Disk Galaxies in the Early Universe: Insights from Cosmological Simulations*, Submitted, 2025 [[2504.01070](#)]
- [9] Zou. S[†] et al. (including **Jinning Liang**), *Disturbed cold gas in galaxy and structure formation*, ApJ **994** (2025) 53 [[2502.14705](#)]
- [10] **Jinning Liang**, F. Jiang[†], H.J. Mo et al., *Connection between galaxy morphology and dark-matter halo structure I: a running threshold for thin discs and size predictors from the dark sector*, MNRAS **541** (2025) 2304 [[2403.14749](#)]
- [11] **Jinning Liang**, F. Jiang[†] et al., *Constrain the Dark Matter Distribution of Ultra-diffuse Galaxies with Globular-Cluster Mass Segregation: A Case Study with NGC5846-UDG1*, ApJ **964** (2024) 53 [[2304.14431](#)]
- [12] **Jinning Liang**, E. Gjergo[†] & X. Fan, *Assessing stellar yields in Galaxy chemical evolution: benchmark on observational stellar abundance patterns*, MNRAS **522** (2023) 863 [[2304.00208](#)]
- [13] E. Gjergo[†] et al. (including **Jinning Liang**), *GalCEM I - A Publicly-Available Detailed Isotopic Chemical Evolution Code*, ApJS **264** (2023) 44 [[2301.02257](#)]
- [14] H. Liu^{*}, **Jinning Liang**^{*} & J. Jia[†], *Deflection and Gravitational lensing of null and timelike signals in the Kiselev black hole spacetime in the weak field limit*, Class. Quantum. Grav **39** (2022) 195013 [[2204.04519](#)]

TEACHING

• PHYS2631 Stars & Galaxies (Undergrad level), Durham University <i>Workshop Demonstrator</i>	Oct 2023 - Apr 2024
• PHYS00407807 Data Analysis in Astronomy (Grad level), Peking University <i>Teaching Assistant</i>	Sep 2026 - Jan 2027

SCIENTIFIC OUTREACH

• Celebrate Science, Durham University <i>Educated children with basic knowledge and illustration about gravitational lensing and galaxy formation</i>	2023
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TALKS (* Denotes invited talk)

• Conference Talk, Kavli Astrophysics Symposium 2026, Cambridge, UK	2026
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AGN in COLIBRE dwarf galaxies

• **Workshop Talk***, University of California Santa Cruz, Santa Cruz, USA 2026

Disk formation in halo conditions

• Seminar Talk, The University of Western Australia, Perth, Australia 2026

Disk formation in halo conditions

• Conference Talk, The 3rd Shanghai Assembly on Cosmology and Structure Formation, Shanghai, China 2025

Connection between galaxy morphology and dark-matter-halo structure

• Conference Talk, CSST annual meeting, Kunming, China 2025

Galaxy-halo coevolution: simulations and mock observations – Halo condition on disc formation

• **Seminar Talk***, The Chinese University at Hongkong, Hongkong, China 2025

Connection between galaxy morphology and dark matter halo structure – Halo condition on disc formation

• Seminar Talk, Flat Talk, Durham, UK 2024

AGN feedback by line-driven winds

• Workshop Talk, 20th Durham-Edinburgh eXtragalactic (DEX) Workshop, Durham, UK 2024

New morphological decomposition method and varying circularity threshold

• **Seminar Talk***, Shanghai Astronomical Observatory, Shanghai, China 2023

Connection between galaxy morphology and dark-matter halo structure

• **Seminar Talk***, Shanghai Jiaotong University, Shanghai, China 2023

Dark matter properties and its connection with galaxy morphology

• Seminar Talk (online), University of Arizona, Arizona, AZ, USA 2022

Globular Clusters in UDGs

• Conference Talk, ISM Physics and Chemistry Conference, Yichang, Hubei, China 2022

Galactic stellar abundance scatter investigated through yield analysis in galaxy chemical evolution

SELECTED HONORS AND AWARDS

• Science and Technology Facilities Council Scholarship 2023

• Yu Gang - Song Xiao Scholarship of Wuhan University 45/30000 2022

• First-class Scholarship of Wuhan University Top 5% 2022

• MCM&ICM Finalist Award Top 2% 2022

• National Astronomical Observatories Scholarship 3/600 2021

MENTORSHIP

• **Haiyang Xing**, postgraduate at Peking University since Jan 2024

Finding Milky Way catalog using galactic decomposition method (MorphDecom), in preparation

• **Peng Xu**, undergraduate at Tsinghua University since Mar 2024

Classifying and visualizing cosmic web with T-web method and DisPerSE

• **Yalin Wu**, undergraduate at Peking University since Oct 2024

Simulating stellar streams and globular clusters in semi-analytical model, in preparation

• **Yiheng Tian**, undergraduate at Peking University Dec 2024 - May 2025

Studying the evolution of bars quantities in IllustrisTNG, undergraduate thesis

• **Jianyuan Luo**, postgraduate at Peking University since Mar 2025

Finding inner mini-disk of massive galaxies at high redshift using cosmological simulations, [2609.27460]

• **Diyuan Wang**, Internship student at Peking University since Apr 2026

Connection between galaxy spheroids and halo conditions, in preparation

SERVICE

• **Organizer (Chair)** of 1st PKU Graduate Forum on Astronomy and Astrophysics 2026

VISITING EXPERIENCES

• International Centre for Radio Astronomy Research, The University of Western Australia Jul 2026

• The Kavli Institute for Astronomy and Astrophysics, Peking University Mar 2023 - Oct 2024

• Shanghai Astronomical Observatory Jan 2022 - Feb 2022

SKILLS

Programming Languages & Software: Python (Extensively), Mathematica (Extensively), Matlab and LaTeX

Language: Mandarin (Native), English (Proficient)

Simulations & Semi-analytical model: *IllustrisTNG*, *EAGLE*, *Swift-COLIBRE*, *NuPyCEE* and *SatGen*

PROFESSIONAL REFERENCES

- **Prof. Fangzhou Jiang**

The Kavli Institute for Astronomy and Astrophysics, Peking University

Email: fangzhou.jiang@pku.edu.cn

- **Prof. Cedric Lacey**

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